LLM / Python Developer Internship Assessment

DO NOT Save This Google Doc Locally, Always Open This On The Browser (Refresh)

TechStack: Python, JS, DataBases, Streamlit, tradingview lightweight charts

Objective

The goal of this assessment is to evaluate your ability to:

- Deal with OHLCV data of financial assets
- Plot and visualize the data in a good looking dashboard created using streamlit or customJS
- Use tradingview lightweight charts to deliver the price visualisation in a streamlit or JS dashboard app, *don't use any other visualization library*
- Query the data and present results from LLM (Gemini) using Agentic AI - Chatbot

Task

1. Use the data from this spreadsheet (TSLA stock data) for your candlestick chart and all other visualizations. File → Download —> csv file
2. Create **arrow markers** green / up pointed arrow when `direction` is LONG and red / down pointed arrow when `direction` is SHORT, when direction is **None**, use a yellow circle. The red arrow needs to be above the candle and green arrow (pointing up) needs to be below the candle
3. Create a **green band for the support prices** , just use the lowest value of the list of prices as the lower bound of the support band and the highest value of the list (in each row of the Support column) as the upper bound of the support band.
4. Create a **red color band for the resistance prices**, just use the lowest value of the list (in each row of the Resistance column) as the lower bound of the resistance band and the highest value of the list (in each row of the resistance column) as the upper bound of the resistance band.
5. **Agentic AI task** - Create a chatbot with gemini API that answers questions about the above spreadsheet data. (Eg: ***How many days in 2023 was TSLA bullish?*** , brownie points for more coming up with more creative and thoughtful template questions about the data in the spreadsheet)

All the above elements need to be overlaid onto a single plot, i.e. markers, bands everything on top of the candle stick chart (see reference image in the next page). Agentic AI chatbot can be presented in a separate tab in the same streamlit app

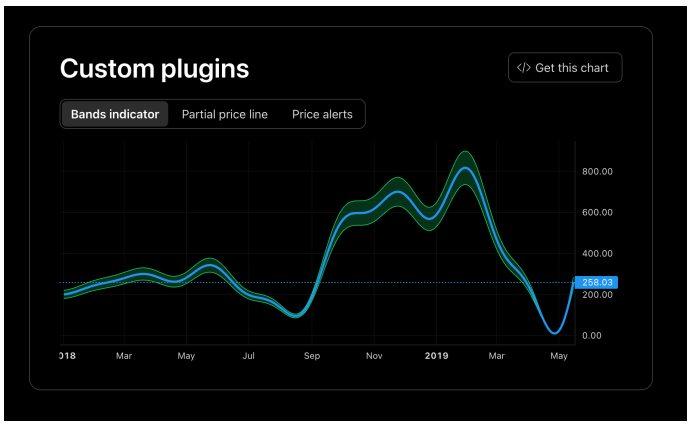
BONUS: Create a moving animation video of the candles and bands and markers being created similar to what we might see in a bar replay in TradingView backtest

Deliverables

- You can design this dashboard in JS or python.
- Deploy the working streamlit app in the streamlit community cloud or if you are using JS (React, Vue etc..) to build the dashboard, deploy it in Heroku or similar .
- Email the deployed link to team@indiancapital.in in the same thread where you confirmed your assessment readiness. *Double check if your deployment works* and make a screen recording with commentary of how to use/navigate your streamlit app and share the .mp4 file via Google drive.
    - Make sure the dashboard URL link you send us is publicly accessible , so we can see it.
- Please keep your code in a PRIVATE Github Repository (no need to share it with indiancapital)

References:

- GitHub - freyastreamlit/streamlit-lightweight-charts
- Lightweight Charts™ library — TradingView (See **Indicators & Markers**, **Custom Plugins**)





Below is a reference image for a candle stick chart with markers and bands overlaid which is the plot expected from your dashboard

